

KARTIK SONI

Software/ML Engineer

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TECHNICAL SKILLS

AI / ML	PyTorch · TensorFlow/Keras · Transformers · HuggingFace · LLMs · Fine-tuning · CNNs · NLP · Computer Vision OpenCV · Naïve Bayes · Cross-validation · Hyperparameter Tuning
Backend	Spring Boot · Java · Python · Django · Node.js · REST APIs · Redis · Microservices · JUnit · Mockito · Swagger
Frontend	Angular · React.js · TypeScript · JavaScript · HTML/CSS · MVVM · Lazy Loading · Reactive Forms
DevOps / Infra	Kubernetes · Docker · GKE · Helm · CI/CD · Jenkins · NGINX · GitHub Actions · PostgreSQL · MongoDB
Practices	System Design · HLD/LLD · Agile/Scrum · Design Patterns · Code Review · Technical Documentation

EXPERIENCE

Software Engineer (Full-Stack, Engineering Lead) · [Whova](#) San Diego, CA, USA | June 2023 - May 2026

- Appointed Engineering Lead for end-to-end product shipment - owned POC coding, MVP scoping, Epic estimation, stakeholder communication, and cross-team delivery for an event management platform serving large-scale bookings.
- Built an AI agent with a tool-calling feedback loop (DB queries, web search, email) on a self-hosted Llama/vLLM/KServe stack deployed on GKE, with capabilities exposed as reusable MCP servers and served via FastAPI.
- Re-architected a high-traffic Angular web app with MVVM, lazy loading to reduce initial load time (from 5 to 2 seconds), OnPush change detection to improve runtime performance significantly.
- Benchmarked Django vs. Spring Boot for new backend microservices using k6; identified Spring Boot's thread-pool model as better for concurrency despite higher baseline resource use, driving the architecture decision.
- Modernised legacy Java monolith into Spring Boot microservices; documented all RESTful APIs with Swagger and achieved >90% unit-test coverage with JUnit and Mockito.
- Containerised and deployed services on Google Kubernetes Engine; authored deployment, service, and virtual-service manifests, cutting release friction for a multi-service stack.
- Recognised as Employee of the Month for engineering innovation; mentored 3 interns through onboarding, Agile practices, and project bootstrapping.

ML Researcher · [UMass Amherst - Prof. Jaime Davila](#) Amherst, MA, USA | Feb 2023 - May 2023

- Designed and trained CNN architectures for medical image classification (disease detection) on datasets of 4 000 - 13 000 images, iterating on kernel sizes, depth, and regularisation to combat overfitting.
- Applied 4-fold cross-validation and systematic hyperparameter search (learning rate, depth, optimiser), switching to SGD and achieving a final F1 score of 91%.
- Improved dataset quality by sourcing diverse, well-distributed images and applying augmentation (rotations, flips, brightness/contrast shifts), boosting generalisation on held-out sets.

Teaching Assistant - Systems Programming · [UMass Amherst](#) Amherst, MA, USA | Sep 2022 - May 2023

- Supported 100+ students on concurrency, synchronisation, pointers, dynamic memory, and control flow; designed assignments and labs that consistently received positive feedback.

Software Engineer Intern · [ISO New England](#) Holyoke, MA, USA | Jun 2022 - Aug 2022

- Migrated a legacy C++ application to a Spring Boot Java microservice; added JWT-based auth, role-based access, and Redis caching, reducing average API latency noticeably.
- Integrated SLF4J + Sentry for structured error logging; applied Lombok and SonarLint to raise code quality; scheduled automated emails via SendGrid cron jobs.

PROJECTS

Multimodal Language Model (Indic LLM) · Python · PyTorch · mBART · Detectron2 · LoRA/QLoRA · Multi-GPU Distributed Training

- Built a 3B+ parameter Indic language model trained on low-resource, morphologically rich datasets, enabling cross-lingual generation.
- Engineered a multi-GPU data-parallel distributed training pipeline with fault-tolerant checkpointing and periodic evaluation, enabling stable large-scale pretraining and reproducible experiment workflows.
- Performed systematic evaluation of tokenization strategies (BPE, WordPiece, SentencePiece) and developed a morphology-aware Indic subword tokenizer that reduced fragmentation in agglutinative structures and improved downstream performance.
- Fine-tuned mBART for multimodal neural machine translation by integrating visual region features extracted via Detectron2 from 30K+ image-text pairs, along with NER-based entity masking to improve named-entity consistency.
- Evaluated architectural and training trade-offs, including early vs. late multimodal fusion and parameter-efficient fine-tuning (LoRA/QLoRA) vs. full fine-tuning, selecting optimal configurations under compute and performance constraints.
- Achieved 40.51 BLEU on multimodal MT, outperforming a text-only mBART baseline (38.84 BLEU), demonstrating gains from visual grounding and optimized training design choices.

Smart Home IoT System · Raspberry Pi · Python · Node.js · Angular · Supabase PostgreSQL

- Built a Raspberry Pi temperature/humidity monitor (DHT11) with sensor data streamed to Supabase PostgreSQL via a Node.js API; deployed Angular dashboard on Vercel for real-time room visualisation and remote relay control.
- Added Pi-hole DNS-level ad blocking and relay-based scheduled lighting control, demonstrating edge AI/IoT system design end-to-end.

Real-Time Mask Detection (CNN) · TensorFlow · Keras · OpenCV · Python

- Trained a CNN on a manually labelled, augmented dataset (rotations, flips, brightness/contrast shifts, zoom) to achieve 92% real-time mask-detection accuracy.
- Deployed as an open-source initiative during COVID using OpenCV for live video inference; contributed to public safety tooling adopted by multiple organisations.

NLP Spam Filter · Python · NLTK · Naïve Bayes · Scikit-learn

- Built an SMS/email spam classifier using NLP pre-processing (tokenisation, stemming, TF-IDF) and a Naïve Bayes model; reached an F1 score of 87% on unseen data.

EDUCATION

B.S. Computer Science with Honors · GPA 3.9/4.0 Aug 2019 - May 2023

University of Massachusetts Amherst

Relevant: AI · Machine Learning · Computer Vision · NLP · Search Engines · Algorithms for Data Science · Reasoning Under Uncertainty · Systems Programming · OS · Networks · Software Engineering

AWARDS & RECOGNITION

Cum Laude · Dean's List (multiple semesters) · Bay State Scholarship (Masters) · Employee of the Month @ Whova